

- **Capacity (GB):** The total amount of memory on the kit. See Section 4.5 for how much you need.
- **Module count:** Two modules (2×16) run in dual-channel mode, giving better bandwidth than a single 32 GB stick. If your motherboard has four slots, use matched pairs.
- **Frequency (MT/s):** Also labelled as speed, it's the number of data transfers per second. Higher MT/s increases bandwidth.
- **CAS Latency (CL) and Timings:** A series like CL30-36-36-76 (often simplified to CL30) describes the delay, in clock cycles, for operations like column read, row precharge and active to precharge. Lower numbers mean less waiting. A kit with high frequency and low latency (e.g., DDR5-6000 CL30) generally performs better than one with higher frequency but higher latency (e.g., DDR5-7200 CL38). In terms of CAS latency, focusing solely on the first number is not enough – because a complete set of timings influences actual performance.



Intel's Gear Modes

New Intel platforms decouple the CPU's internal memory controller (IMC) from the memory clock using Gear modes. Gear 1 runs the IMC at the same frequency as the memory, Gear 2 at half speed and Gear 4 at one quarter. Higher gears enable extreme memory speeds but increase latency. Benchmarks show that DDR5-8800 running in Gear 2 can outperform DDR5-9600 in Gear 4 because the lower gear delivers lower latency. For gaming, a balanced frequency like DDR5-6000 or 6400 in Gear 2 often yields the best combination of bandwidth and latency. Gear modes are selected automatically on most motherboards; manual tuning is for enthusiasts.

XMP and EXPO: Unlocking rated performance

Out of the box, your memory will run at conservative JEDEC defaults—typically DDR5-4800 CL40. To use the high-performance settings printed on the box, you must enable a memory overclocking profile:

- **Intel XMP (Extreme Memory Profile):** Developed by Intel; sets frequency, timings and voltage for modules validated on Intel platforms.
- **AMD EXPO (Extended Profiles for Overclocking):** Designed for AMD Ryzen systems; conceptually the same as XMP but tailored for AMD CPUs.

Most modern motherboards support both XMP and EXPO, and many memory kits embed both profiles. For example, Corsair's Vengeance DDR5-6000 CL30 kit has XMP and EXPO profiles to ensure optimal performance on Intel 15th Gen or AMD Ryzen 9000 systems.

Benefits of Using XMP/EXPO

Enabling XMP/EXPO applies manufacturer-tested timings and voltage values, boosting performance without manual tuning. XDA Developers note that a DDR5-5600 CL36 kit running at default settings (4800 CL40) exhibits noticeably higher latency and slower frame pacing; turning on XMP/EXPO reduces memory latency and improves smoothness in games. The process is reliable because the profile is validated by the memory manufacturer.

How much RAM do you need?

Memory requirements vary widely. Choosing the right capacity ensures that your money is spent where it matters.

- **8 GB:** Suitable only for basic browsing, email and very light office tasks. Modern operating systems can use most of this just to run; heavy multitasking will cause swapping to disk.
- **16 GB:** The current baseline for mainstream PCs. It handles office work, web browsing with numerous tabs, and casual gaming, but may struggle with modern AAA titles or professional software with large datasets.
- **32 GB:** The new sweet spot for gamers, content creators and engineers. Corsair's Digital Foundry testing found that 32 GB DDR5-6000 CL30 delivered up to 14 % higher average frame rates compared with 16 GB DDR5-4800. It also provides headroom for streaming, running virtual machines and multi-tasking.
- **64 GB and beyond:** Recommended for professionals working with 4K/8K video editing, 3D modelling, AI/ML workloads or large scientific datasets. Crucial notes that top-end users such as engineers or video editors need 64 GB or more to avoid bottlenecks. For gamers who stream and run resource-intensive mods or for those building AI PCs, 64 GB ensures future-proofing.

When planning your build, consider your GPU and CPU. Pairing a high-end GPU (e.g., NVIDIA GeForce RTX 5000 or AMD Radeon RX 8000) with too little RAM will limit performance. For an Intel Core 15th Gen i7 or AMD Ryzen 9 9000 CPU, 32 GB DDR5-6000 CL30 or 6400 CL32 is ideal. If you anticipate heavy content creation or AI workloads, step up to 64 GB.